

CNS Nicotinic Receptors

Possible Therapeutic Targets in Neurodegenerative Disorders

Jennifer A. Court and Elaine K. Perry

Medical Research Council Neurochemical Pathology Unit, Newcastle General Hospital,
Newcastle upon Tyne, England

Contents

Summary	216
1. CNS Nicotinic Receptors	217
1.1 Distribution of CNS Nicotinic Receptors	217
1.2 Nicotinic Receptor Subtype Pharmacology and Function	221
2. Receptor Changes	222
2.1 Development and Aging	222
2.2 Degenerative Dementia	223
2.3 Other Neurodegenerative Diseases	223
3. Effects of Nicotine on the Human Brain	224
3.1 Smoking and Parkinson's Disease	225
3.2 Smoking and Alzheimer's Disease	226
3.3 Therapeutic Trials of Nicotine	226
4. Spectrum of Nicotinic Compounds	228
5. Models for the Assessment of Nicotinic Drugs	229
6. Conclusion	229

Summary

CNS nicotinic acetylcholine receptors are a family of ligand-gated cation channels that allow the passage of calcium ions. They are structurally and functionally distinct from nicotinic receptors at the neuromuscular junction. CNS nicotinic receptors show a variation in distribution and pharmacology.

The number of these receptors is reduced with normal aging, and a further reduction in number occurs in dementing conditions that are common in elderly individuals. A comparison of receptor loss in the substantia nigra of patients with Parkinson's disease and Lewy body dementia indicates that this process may occur early in the disease and precede cell loss. In addition, nicotine binding sites in humans are concentrated in the entorhinal cortex and subicular formation, areas that are affected early in the course of Alzheimer-type disease.

The number of nicotine binding sites is high in human neonatal brain, at which stage exposure to nicotine and tobacco smoke is deleterious. Conversely, epidemiological evidence suggests that later in life tobacco smoking may offer some protection against Parkinson's and Alzheimer's diseases. Exposure to nicotinic agonists and tobacco smoke increases the number of brain nicotinic receptors both in humans and experimental animals, although the functional significance of this and its precise relationship to receptor desensitisation is not completely under-

stood. Acute exposure to nicotine and nicotinic antagonists has beneficial and adverse effects, respectively, on cognitive function. Subcutaneous administration of nicotine to patients with Alzheimer's disease improves discriminative sensitivity and reaction times. Available nicotinic agonists appear to induce diverse functional responses, and models for the testing of such novel compounds are needed.

1. CNS Nicotinic Receptors

CNS nicotinic acetylcholine receptors are a family of ligand-gated cation channels with a generally pentameric structure, comprised two α and three β subunits.^[1-4] Electrophysiological and radioligand binding studies have indicated the heterogeneity of CNS nicotinic receptors, with the different receptor subtypes having varying agonist and antagonist sensitivities.^[5-10] These findings have been confirmed by recent molecular biology studies (for reviews see Heinemann et al.,^[1] Deneris et al.,^[2] Sargent^[9] and Role^[11]). In the chick and rodent, at least seven α (α_2 , α_3 , α_4 , α_5 , α_6 , α_7 , α_8) and three β (β_2 , β_3 , β_4) subunits have been identified. Different subunit composition appears to result in distinct, but not exclusive, receptor pharmacology.^[1,2,9,11]

CNS nicotinic receptors are structurally and functionally different from nicotinic receptors at the neuromuscular junction. The major distinction is the high affinity of muscle receptors for the snake (*Bungaris multicinctus*) venom toxin, α -bungarotoxin. This effect is not shared by the majority of nicotinic receptors in the CNS. Furthermore, there is an absence of γ , δ and ϵ subunits in CNS receptors.

There are receptors with high affinity for α -bungarotoxin in the brain. However, the pattern of binding does not parallel that of acetylcholine (in the presence of atropine to block muscarinic receptors), and α -bungarotoxin does not generally block responses to CNS nicotinic cholinergic receptor stimulation. It is receptors with high affinity for nicotine that reflect acetylcholine binding in terms of kinetics, pharmacological profile and distribution.^[12-15] Thus, α -bungarotoxin binding sites seem to be associated with different receptor subtypes.

The major brain nicotinic receptor subtype to

which acetylcholine, nicotine and other agonists such as methyl carbamylcholine and cytosine (fig. 1) bind with nanomolar affinity is composed of α_4 and β_2 subunits.^[16-19] However, α_2 , α_3 and β_4 subunits, although not as prevalent as α_4 and β_2 , may also contribute to brain nicotinic cholinergic receptor activity.^[1,20,21] α -Bungarotoxin binds to α_7 and α_8 subunits.^[4] In the absence of other α or β subunits, α_7 and α_8 subunits are able to form active cation channels that are blocked by α -bungarotoxin.^[22-26]

CNS α -bungarotoxin binding sites will be subsequently referred to as α BTRs, while the binding sites with high affinity for nicotine and other agonists (presumed to be composed of two α and three β subunits) will be referred to as nAChRs.

1.1 Distribution of CNS Nicotinic Receptors

1.1.1 Rodent Studies

The distribution of nAChRs and α BTRs and subunit messenger RNAs (mRNAs) has been studied extensively in rodents by autoradiography and *in situ* hybridisation.

In rats, the highest levels of nAChRs occur in the interpeduncular nucleus, most thalamic nuclei, the superior colliculus and the medial habenula. [³H]Nicotine binding was also prominent in the substantia nigra (pars compacta), ventral tegmental area, selected areas of the hippocampal formation (dentate molecular layer, the subiculum and pre-subiculum), cerebral cortex (laminae I, II and IV) and septum (bed nucleus stria terminalis). Moderate levels of binding were found in the neostriatum, ventral striatum, dorsal tegmental nucleus and cerebellum.^[14]

The pattern of nicotine binding in mice shows some differences to that in rats, with the highest

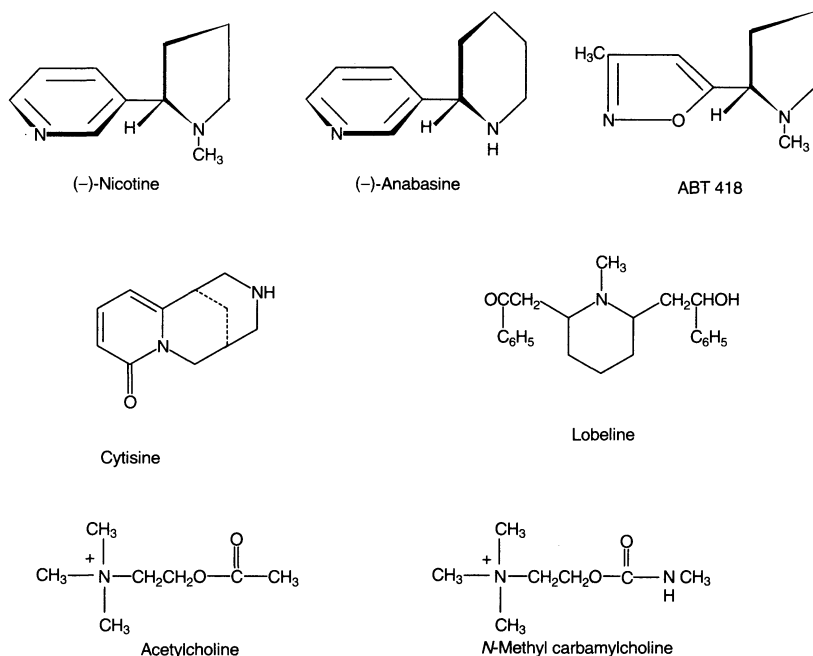


Fig. 1. Structural formulae of nicotinic agonists

levels in the interpeduncular nucleus and relatively lower levels in other brain regions.^[27-29]

The distribution of α_4 and β_2 subunits, as assessed by measuring subunit mRNA, in rodents generally reflects that of [³H]nicotine binding, but other putative nicotinic subunits have more circumscribed patterns of distribution.^[30] The α_3 gene was highly expressed and represented the major α subunit in the medial habenula, layer IV of the cerebral cortex and medial area II of the entorhinal cortex. It is also localised in areas of the thalamus, hypothalamus and brain stem.^[29,30] The α_2 subunits are concentrated in certain interpeduncular nuclei, the dorsal tegmental nucleus, inferior colliculus, parabrachial nucleus and select areas of the hippocampal formation (pyramidal and striatum radiatum of the subiculum and striatum oriens of the CA1).^[29,30] Subunits of the α_5 subtype are detected in distinct layers of the subicular complex (e.g. pyramidal layer), substantia nigra (pars compacta and

ventral tegmental areas), interpeduncular nucleus and dorsal motor nucleus of the vagus nerve.^[31]

β_3 and β_4 subunits were most highly expressed in the medial habenula.^[2,32] In addition, β_4 subunits are distributed throughout the isocortex, the forebrain olfactory regions and the hippocampus formation. An intense signal is displayed in the interpeduncular nucleus and the trigeminal motor V nucleus, and more moderate signals in the amygdala-hippocampal tissue, locus coeruleus, pontine nucleus, supramammillary nucleus of the hypothalamus, septal nucleus and the Purkinje cell layer of the cerebellum.

¹²⁵I- α -Bungarotoxin binding in rats closely parallels the expression of α_7 mRNA, but is distinct from the distribution of high affinity nicotine binding.^[14,24] It is present in the cerebral cortex (highest intensity in laminae I and IV), endopiriform nucleus, hypothalamus, hippocampus (dense labelling in the CA4 region), certain brain stem

nuclei (including the superior and inferior colliculi, locus coeruleus, dorsal tegmental nucleus and interpeduncular nucleus) and amygdala. With the exception of the lateral and medial geniculate nuclei, there was little or no detectable ^{125}I - α -bungarotoxin binding or expression of $\alpha 7$ mRNA in the thalamus, substantia nigra or striatum.

1.1.2 Human Studies

In human brains, complete topographical analysis of nAChR and α BTR distribution has not been achieved. However, autoradiographic and membrane binding studies completed to date indicate broad agreement with rodent studies, but some distinctions are apparent. The highest levels of nAChRs were present in the thalamus, in particular the lateral geniculate nucleus. The principal anterior, dorsomedial and medial geniculate nuclei showed slightly lower levels of binding, and the pulvinar, dorsolateral, reticular and ventral group nuclei demonstrated moderate binding.^[33-35]

In contrast to findings in rats and mice, ^3H nicotine binding in human striatum and the pars compacta of the substantia nigra was almost as high as that in the dorsomedial thalamus (see fig. 2).^[33,34,36] Although lower levels of nAChRs were generally observed in the neo- and archicortex, a distinct pattern of distribution and some areas of high intensity have been reported.^[37] Binding was greatest in the subicular complex, particularly the presubiculum, where it was prominent in the parvo- and magnocellular islands and the middle layer of the entorhinal cortex (fig. 2). There was generally little binding in the hippocampus, with the exception of the striatum lacunosum moleculare in the CA2-3 and, to a lesser extent, the supra- and subgranular zones of the dentate (fig. 2).

In the neocortex, there were distinct patterns of binding in different areas. For example, in the somatosensory and occipital visual cortex, binding was highest in the upper and lower layers, and relatively sparse in layer IV. In the temporal and cingulate cortex, a distinct band was apparent in layer III.^[37]

The distribution of α BTRs and $\alpha 7$ activity in the human hippocampus was, in contrast to nAChRs, concentrated in the dentate and CA1-3 regions,

with little binding in the subicular complex or entorhinal cortex (Rubboli et al., unpublished data).

1.1.3 Correlations Between Receptor Location and Neuronal Projections

The distribution of receptor subclasses and receptor subunit mRNAs does not exactly correspond to any known cholinergic projection. However, nicotine binding concentrates in areas such as the thalamus, superior colliculus and substantia nigra pars compacta (which is also high in $\alpha 4$ mRNA). This is suggestive of a concentration of nAChRs in neurons immediately postsynaptic to the brain stem Ch5 and Ch6 cholinergic projections. These projections are part of the reticular activating system. They are likely to play a major role in arousal and, together with the modulation of thalamic control of sensory input, also in selective attention.

The high density of nicotine binding and $\alpha 4$ subunits in the substantia nigra pars compacta is compatible with a role for nAChRs in controlling dopaminergic projections to the striatum. The relatively intensity of nAChRs in the striatum in humans, together with the well described presynaptic control of dopamine release by nicotinic agonists in the striatum *in vitro*,^[38-40] further indicates a significant role for nAChRs in modulation of the extrapyramidal motor system.

In the hippocampus, the concentrations of cholinergic processes are not reflected in high nicotinic binding site density. However, the concentration of binding sites in the subicular complex suggests the nAChRs are associated with the perforant pathway. This is an excitatory pathway (possibly using glutamate as a neurotransmitter) that projects from the subiculum to the granule cells of the dentate gyrus. The effects of nicotine on learning may be mediated at this locus.

In the neocortex, cholinergic axons are generally more dense in the superficial layers than in the deeper layers, and this tends to be the case with nicotine binding sites. Nicotinic stimulation of isolated mouse cortical synaptosomes has been shown to stimulate the release of ^3H acetylcholine. This indicates the presence of nAChRs on presynaptic

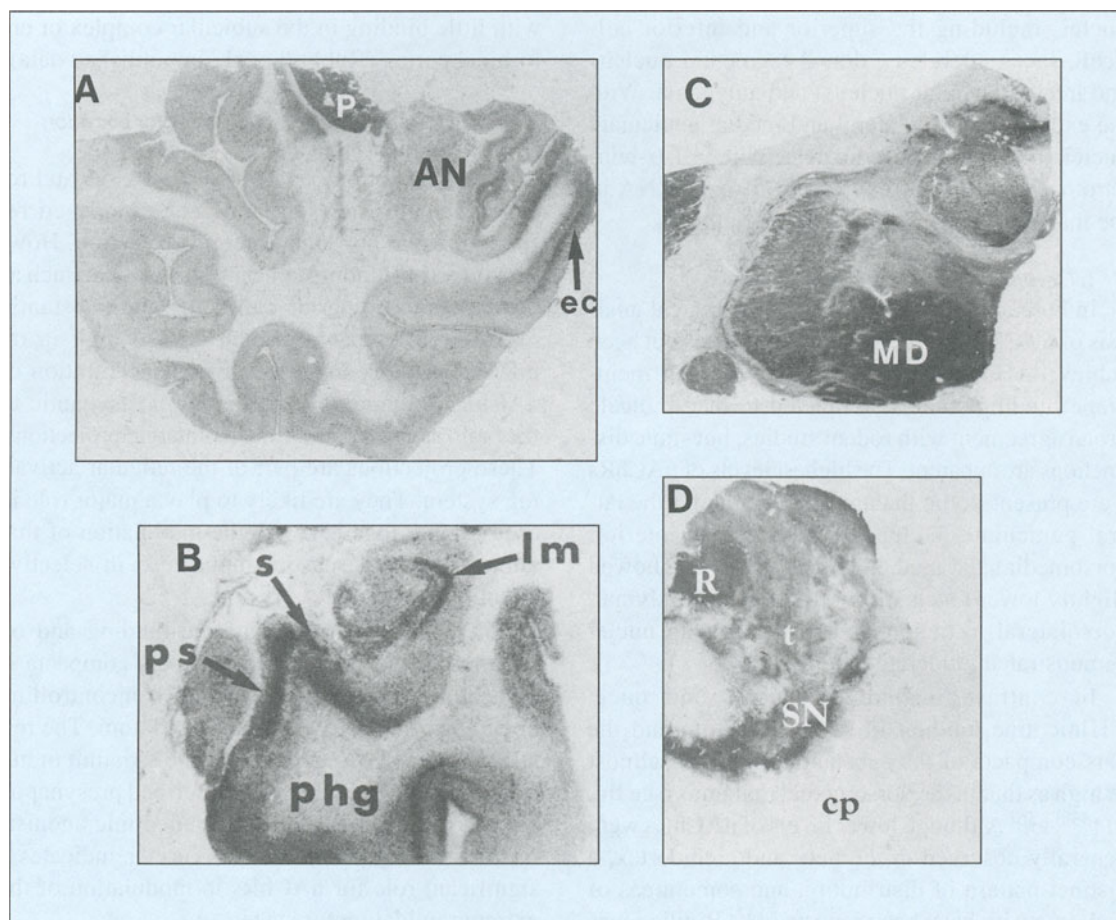


Fig. 2. Distribution of [^3H]nicotine binding (measured at a concentration of 3 nmol/L) in brain sections from adult humans. A. Anterior temporal lobe, including entorhinal cortex; B. hippocampal formation and adjacent parahippocampal gyrus; C. anterior thalamus; D. midbrain at the level of the substantia nigra pars compacta. Abbreviations: AN = amygdala; cp = cerebral peduncle; ec = entorhinal cortex; lm = stratum lacunosum moleculare; MD = mediodorsal nucleus of thalamus; P = putamen; phg = parahippocampal gyrus; ps = presubiculum; R = dorsal raphe nucleus; s = subiculum; SN = substantia nigra pars compacta; t = dorsal tegmentum.

terminals in the cerebral cortex.^[41] However, these terminals may not be projections from the basal forebrain, since lesions in this area have not generally been shown to reduce the number of neo- nor archicortical nAChRs.^[42,43] In contrast, lesions of the thalamic or lateral geniculate projections reduced the number of nAChRs in the corresponding cortical areas.^[44,45] This further establishes a role of nAChRs in the thalamic/reticular activating system modulation of sensory information.

nAChRs are also likely to function postsynaptically within the cerebral cortex, since specific nAChR immuno-staining has been observed in rat and human cortex on pyramidal and other neurons. The staining was associated with dendrites not axons.^[46,47]

The distribution of αBTRs and $\alpha 7$ subunits tends to be concentrated in the neo- and archicortex, and in the latter broadly corresponding to acetylcholinesterase reactivity.^[37] This indicates an association with Ch1-4 cholinergic projections, rather than

Ch5 and 6 projections. The co-existence of both receptor and subunit mRNA in rats and humans suggests a postsynaptic location for these receptors. This location is probably associated with afferent cholinergic input, since intrinsic cortical cholinergic neurons have not been observed in humans. However, the low potency of acetylcholine at this receptor subtype does suggest the possibility of some other endogenous ligand for the receptor. Recently, it has been demonstrated that the endogenous polypeptide thymopoietin is a potent displacer of [125 I]- α -bungarotoxin binding in rat brain.^[48]

1.2 Nicotinic Receptor Subtype Pharmacology and Function

Expression of different subunit combinations in oocytes has allowed the direct comparison of the properties of putative receptor subtypes. Results clearly demonstrate considerable differences in terms of agonist and antagonist sensitivities and rates of desensitisation that occur within seconds of exposure to an agonist (for review see Role^[11]). Thus, a combination of $\alpha 4 \beta 2$ is most sensitive to acetylcholine and (–)-nicotine [concentration of both agents that induces a 50% response (EC_{50}) for chick subunits = 0.7 μ mol/L]. This receptor subtype is partially blocked by another *Bungaris multicinctus* venom toxin, neuronal bungarotoxin, but not by α -bungarotoxin, and has a relatively slow desensitisation rate. A subunit combination of $\alpha 3 \beta 2$ has a lower sensitivity to acetylcholine (EC_{50} for chick subunits = 5 μ mol/L) and (–)-nicotine, is blocked by neuronal bungarotoxin but not α -bungarotoxin, and desensitises rapidly.

In contrast, $\alpha 7$ -containing receptors are blocked by both α - and neuronal bungarotoxin, are relatively insensitive to acetylcholine (EC_{50} for chick subunits = 115 μ mol/L), slightly more sensitive to (–)-nicotine (EC_{50} for chick subunits = 10 μ mol/L) and desensitise very rapidly. Although agonist binding occurs at the α subunit, it has become clear that β subunit composition also influences receptor ligand specificity and desensitisation rates at nAChRs.^[1,20,49,50]

It had been assumed, by analogy with the neuromuscular receptor, that CNS nAChRs mainly gate sodium/potassium (Na^+/K^+) currents. Therefore, receptor stimulation would be expected to reduce membrane potential and facilitate the generation of action potentials. However, it is now becoming evident that neuronal nAChRs (medial habenula neurons, $\alpha 3 \beta 4$ expressed in oocytes, superior cervical ganglion cells and myenteric plexus neurons) and α BTRs exhibit a significant permeability to calcium ions.^[51–56] Neuronal nAChRs and α BTRs, therefore, represent a mechanism of increasing intracellular calcium. This action is largely independent of membrane potential [unlike voltage-sensitive calcium channels and *N*-methyl-D-aspartate (NMDA) glutamate receptors]. Also, because of the relatively low tissue density of the receptors this action would be unlikely to give rise to calcium-mediated neurotoxicity (the concentration of nAChRs is generally 1 or 2 orders of magnitude lower than that of NMDA receptors). Because of the ability to modulate calcium levels, the potential importance of nicotine receptors in neuronal cell growth and development is now being recognised.^[57,58]

A particularly interesting feature of brain nAChRs, and one of possible therapeutic significance, is the upregulation of this receptor in response to exposure to tobacco smoke, nicotine and certain other nicotinic agonists [e.g. anabasine, cytisine and methyl carbamylcholine (fig. 1)].^[27,29,59–62] This phenomenon appears to be specific to brain nAChRs^[63] and to those of the $\alpha 4 \beta 2$ subtype,^[19] and does not occur equally across all brain areas.^[27,62] It is controversial whether the increase in receptor number is associated with an increase in functional receptors or merely results in a greater number of desensitised receptors. Clearly, the relationship between agonist-induced tolerance and receptor upregulation is complex.^[28,64] However, it may be that both can occur depending on the dose of the agonist and the duration and frequency of exposure. Indeed, it has been demonstrated that exposure to an agonist *in vivo* can lead to subsequent increases in nicotinic response measured *in vitro*,^[65] and nicotine tolerance *in vivo* caused by nicotine exposure does not

persist as long as the elevated number of brain nAChRs.^[66]

2. Receptor Changes

The density of nicotinic receptors in the human brain (as estimated from the number of [³H]nicotine binding sites) is low compared with other 'classical' neurotransmitter receptors. However, in contrast to many of the other receptors, there have been consistent reports of substantial changes in nicotinic receptors with aging (post-maturity) and disease. These changes invariably consist of a reduction in the number, but not the affinity, of the receptors. The changes appear to occur at early stages of neuropathological changes associated with the disease, and exceed (and presumably precede) neuron loss.

Recent evidence suggests a role for nicotine receptors in governing the movement of growth cone filopodia via oriented intracellular calcium gradients.^[58] This indicates the potential of nicotinic acetylcholine interactions in shaping brain structure and function. This effect may continue throughout life.

Nicotinic receptor malfunction due to aging or disease is likely to be associated with diminishing the entry of calcium ions into neurons. This may lead to long term impairment of calcium-dependent processes, such as protein phosphorylation, and thus to neurodegeneration. It is unlikely that in adults with neurodegeneration, neuronal death has occurred through overstimulation and excessive calcium entry, as is the case with the stimulation of excitotoxic NMDA receptor calcium ionophores. The level of intracellular calcium directly associated with neurodegeneration^[67] is many orders of magnitude (> 4) in excess of the normal level. On the other hand, insufficient calcium levels can also lead to neurodegeneration. Furthermore, the paucity of neuronal nicotinic receptors in the adult brain together with extremely rapid desensitisation mechanisms are inconsistent with an excitotoxic scenario of neurodegeneration.

2.1 Development and Aging

Since aging begins at conception, a complete description of fluctuations in nicotinic receptors

would ideally cover the entire lifespan. However, reports of receptor distribution and density in fetal, neonate and adolescent human brains are sparse. Nevertheless, studies have shown that intense nicotine but not α -bungarotoxin binding occurs in the fetal brain stem and cerebellar dentate nucleus^[68] (Court et al., unpublished data). These levels of binding are above those found in adults. In frontal cortex and hippocampus, there are transient perinatal peaks in nicotine binding^[69] (Court et al., unpublished data).

The vulnerability of specific brain areas, especially in the brain stem, to pre- or postnatal trauma/anoxia may relate to the over-expression of nicotinic receptors.^[68] Certainly, prenatal exposure to maternal tobacco smoking is considered detrimental to normal human brain development, and is consistent with the adverse effects of nicotine on the development of cholinergic markers in rat brains.^[70] Therefore, nicotinic antagonists would be expected to have therapeutically relevant neuroprotective actions in developing human brains. It is curious that nicotinic agonists (such as nicotine) behave like 'time-lapsed' antagonists via their ability to rapidly desensitise nicotinic receptors in mature brains. It would be interesting to discover if perinatal receptors differ from those in the adult in this respect.

Following development there appears to be a relentless decline in the number of nicotinic receptors in those human brain areas that have been investigated to date. The data are remarkably consistent, with approximately 50% reductions in nicotine binding in the hippocampus and neocortex^[69,71] and in immunoreactivity to the nicotinic monoclonal antibody WF6 in the frontal cortex^[72] from adulthood (20 to 40 years) to the tenth decade of life.

In the medial temporal lobe, the loss is most extensive in the entorhinal cortex and less extensive in the dentate fascia or hippocampal formation. This coincides with the relative vulnerability of the entorhinal cortex to Alzheimer's-type pathology, such as β -amyloid plaques or neurofibrillary tangles (Court et al., unpublished data). There are also age-related reductions in α -bungarotoxin

binding in the entorhinal cortex (Court et al., unpublished data). In one study comparing different brain regions, the entorhinal cortex and pre-subiculum were shown to be the regions with the highest levels of nicotine binding in the human cortex.^[37] It has been suggested that this differential distribution and vulnerability may relate to the particular function of the entorhinal cortex. The area is a highly active multimodal signalling centre that simultaneously receives information from numerous different regions, especially the association cortex.

2.2 Degenerative Dementia

Numerous reports^[71,73-83] indicate the importance of the loss of cortical nicotinic receptors in Alzheimer's disease. In patients with this disease, there is a reduction in the binding of nicotine, methyl carbamylcholine and acetylcholine (the latter in the presence of atropine to block muscarinic receptors), but not in α -bungarotoxin. This is consistent with an involvement of the predominant $\alpha 4$ subtype, although the human gene for this molecule has not yet been cloned. The change is recorded throughout the archi- and neocortical areas examined, and relates to the severity of dementia. It is not generally observed in subcortical regions, such as thalamus and striatum,^[33,35,74] that are not considered to be directly involved in dementia.

Whether the cortical involvement relates primarily to the loss or dysfunction of cortical input systems, such as the cholinergic projection from the basal forebrain or those from the thalamus, or is associated with pathology of cortical intrinsic neurons is not yet clear. Immunocytochemical studies indicate a reduction in receptor-bearing cortical neurons in Alzheimer's disease^[72] and there is evidence for receptor localisation on both forebrain and thalamic input systems.^[36] This suggests a possible contribution from all of these pathways to the pathological change associated with the disease.

Significant correlations between the loss of the cholinergic marker enzyme choline acetyltransferase (ChAT) and nicotine binding in Alzheimer's

disease have been noted. However, it is not clear how the 2 changes are related. To judge by human cortical data in normal aging, receptor loss precedes degeneration of the cholinergic input. For example, reductions in nicotine receptor binding in the frontal cortex are not accompanied by changes in ChAT, and in the entorhinal cortex the receptor loss occurs earlier than reductions in ChAT (after 20 compared with 50 years, respectively).^[69,84]

It seems likely that the cortical receptor population is predominantly postsynaptic to the cholinergic input. Thus, the early changes in these receptors may reflect direct modulation resulting from dysfunctional changes in the cholinergic input or – independent of acetylcholine – primary alterations in intrinsic cortical neurons.

Whichever of these occurs, the failure of cholinergic therapy aimed specifically at muscarinic receptors in Alzheimer's disease could be due to the necessity to target nicotinic, in addition to muscarinic, receptors. Long term consequences of decreased nicotinic receptor function in terms of impairing calcium influx are likely to involve degeneration of receptor-bearing neurons. This again provides a possible explanation for the inadequacy of current cholinergic therapy. In Parkinson's disease, the success of transmitter replacement therapy may be related to the integrity of nicotinic activity in striatal target areas (see section 2.3). Interestingly, although the anticholinesterase tacrine does not result in a substantial symptomatic improvement in many patients with Alzheimer's disease, it can offset further deterioration in the majority of patients.^[85]

2.3 Other Neurodegenerative Diseases

Changes in nicotinic receptors have also been identified in other forms of dementia,^[74,78-80] including that related to Parkinson's disease, Lewy body dementia (variously referred to as dementia of Lewy body type, Lewy body variant of Alzheimer's disease or diffuse Lewy body disease) and progressive supranuclear palsy. In Parkinson's disease and Lewy body dementia, receptor loss occurs more in the neo- than archicortex and is more

extensive in demented than nondemented Parkinsonian patients^[33,79] (Perry et al., unpublished data). In Alzheimer's disease, as in normal aging, the loss of nicotinic receptors is most marked in the entorhinal cortex compared with other hippocampal regions (Perry et al., unpublished data), although normal age-associated changes in this area complicate interpretation of quantitative analyses of elderly patients with the disease.

The most clinically significant degenerative change in Parkinson's disease (i.e. loss of dopaminergic neurons in the substantia nigra pars compacta) is accompanied by a decrease in the normally relatively high density of nicotinic receptors in this region (Perry et al., unpublished data). The receptor loss in the lateral portion of the nucleus parallels the extensive reduction in neuron number both in terms of the distribution and density. Therefore, it is likely that the receptor is primarily localised on the dopaminergic neurons that receive a cholinergic input from the Ch5 and Ch6 cell groups in the brain stem. This is consistent with the high density of α_4 mRNA in rodent substantia nigra (section 1.1).

Evidence that the loss of nicotinic receptors may precede neurodegeneration of dopaminergic A9 neurons, stems from analysis of patients with Lewy body dementia. In this disease, neuronal loss is only moderate (20 to 30%) and so there is absent or only mild extrapyramidal symptoms. However, nicotine binding in the lateral portion of the pars compacta is decreased to a greater extent than the neuronal loss (over 60%). Curiously, in the striatum, where in rodents there is little or no α_4 mRNA, the level of nicotine binding is not substantially changed in Parkinson's disease^[86] or Lewy body dementia (unpublished observations). This can be interpreted in several different ways: (i) as evidence of compensatory axonal terminal sprouting in surviving neurons; (ii) the additional localisation of receptors in other striatal inputs; or (iii) as a species difference in receptor localisation between rodents and humans.

In Parkinson's disease and Lewy body dementia, an area in the dorsal tegmentum corresponding to the position of the Ch5 cholinergic neurons is also

affected by moderate reductions in nicotinic receptors (Perry et al., unpublished data). Although the clinical correlates of this effect are unknown, an alteration in rapid eye movement (REM) sleep is a likely consequence of pathology in this area.

Nicotinic receptors have not yet been examined in a range of other CNS disorders affecting cortical (e.g. Pick's or Creutzfeldt-Jakob diseases), subcortical (e.g. Huntington's chorea or striatonigral degeneration), cerebellar (e.g. olivopontine cerebellar ataxias) or spinal cord (e.g. motoneuron disease) regions. Investigation of patients with Huntington's chorea would help to answer the question of whether human striatal interneurons (which degenerate in this disease) are nicotinic receptor positive. Although there is only a relatively low density of nicotine binding in the ventral horn of the human spinal cord,^[36] it would be interesting to establish whether early alterations in the receptor are associated with motoneuron degeneration. This is of particular interest since this disease shares some of the clinical and pathological features of Alzheimer's and Parkinson's disease in the dementia complex of Guam – a disease involving impairment of movement and cognition that is unique to this island.

3. Effects of Nicotine on the Human Brain

Nicotine was originally tested for therapeutic effects on the extrapyramidal symptoms of Parkinson's disease in the 1920s^[87] and more recently for the cognitive dysfunction associated with Alzheimer's disease. Paradoxically, the motor disorder observed in Parkinson's disease is related to a relative hyperactivity of cholinergic function in the striatum whereas the cognitive impairment of Alzheimer's disease involves decreased cortical cholinergic activity.

During the last decade, CNS nicotinic receptor subtypes (primarily α_4) with a high affinity for acetylcholine and nicotine have been identified. This finding has spurred the contemporary development and application of therapeutic CNS-active nicotinic agents. In addition, epidemiological re-

search has indicated an inverse relationship between tobacco smoking and the incidence of Parkinson's and Alzheimer's diseases (see sections 3.1 and 3.2). This has fostered the notion that nicotine may be beneficial in certain CNS diseases.

3.1 Smoking and Parkinson's Disease

Almost all investigations of the relationship between smoking habits and the incidence of Parkinson's disease have provided evidence that Parkinsonian patients are less likely to be tobacco smokers than aged-matched normal individuals, i.e. smoking appears to provide protection against the development of the disease. Early reports to this effect date back to the 1960s (e.g. Nefzger et al.^[88]). Most of the more recent epidemiological data support these original findings:

- 63% of patients with Parkinson's disease never smoked compared with 47% of matched controls (defined as healthy individuals without Parkinson's disease) [$p < 0.0001$]^[89]
- 26% of patients with Parkinson's disease had a history of smoking compared with 33% of controls^[90]
- A history of smoking up to 20 years earlier was associated with a risk of developing Parkinson's disease that was around half of that in non-smokers^[91]
- Fewer Parkinsonian patients (7%) ever smoked compared with controls (39%) and many patients with Parkinson's disease (82%) stopped smoking before the onset of the disease^[92]
- The relative risk of smokers having Parkinson's disease is approximately 0.5 that of non-smokers.^[93]

While most investigators agree on the inverse relationship between smoking and Parkinson's disease, interpretations of the nature of the relationship are extraordinarily diverse. These range from fundamental mechanisms of neuroprotection to incidental differences in personality. For example, it has been claimed that there is a typical premorbid personality in Parkinson's disease. Patients are described as moralistic, law abiding, conscientious

and adverse to taking risks.^[94] An alternative explanation in this context is that declining numbers of nicotinic receptors (evident in midbrain and cortex) and decreasing dopamine levels associated with early stages of the disease, reduce the urge to smoke. Such prodromal pathology could even account for subtle personality changes seen in patients with the disease.

Addiction to nicotine has been linked to high levels of cerebral dopamine.^[95] Moreover, since nicotine is associated with the release of striatal dopamine, smoking may postpone the development of clinical symptoms in Parkinson's disease by virtue of elevating synaptic dopamine levels.

There are several neuroprotective theories of smoking and Parkinson's disease:

- Activation of the nicotinic calcium channels has neurotrophic effects^[58]
- Increased melanin in smokers acts as a neurotoxic 'sink',^[96] whereby the pigment, by binding endogenous or exogenous toxins, removes the toxins from the cytosol
- Cytochrome P450 induced by the aromatic hydrocarbons in tobacco smoke promotes detoxification of toxic agents that may be implicated in the development of the disease^[97]
- Hydrazine in the smoke protects dopaminergic nigrostriatal neurons from 1-methyl-4-phenyl-1,2,3,6-tetrahydropyridine (MPTP)-type damage.^[98]

In experimental animals, nicotine promotes dopamine release and protects against the degenerative changes associated with unilateral lesioning of the nigrostriatal pathway^[99-101] or, more controversially, MPTP toxicity.^[102] Nicotine also attenuates the loss of neocortical neurons, including cholinergic neurons, following destruction of the ascending basal forebrain.^[103]

In addition to the long term effects of smoking on the course of Parkinson's disease, there are also immediate symptomatic benefits. Reductions in tremor, rigidity, bradykinesia and frozen gait were recently observed in 6 patients after they had smoked a cigarette.^[104] The effects lasted 10 to 30 minutes. These beneficial effects were not as

marked with nicotine chewing gum. This indicates either that optimal therapeutic effects of nicotine are obtained with intermittent as opposed to continuous delivery, or that some other component of tobacco smoke is beneficial.

3.2 Smoking and Alzheimer's Disease

The relationship between smoking and Alzheimer's disease has not been examined to the same extent as that with Parkinson's disease. Although not all studies have noted an inverse correlation (in fact, the reverse in one instance^[105]) there are a few rather convincing observations. Van Duijn and Hofman^[106] examined 193 patients with familial Alzheimer's disease and found a relative (dose dependent) risk of developing dementia of 0.35 in smokers compared with nonsmokers. Most impressively, the mean age of clinical onset was 4.17 years later in smoking compared with non-smoking members of 6 families. Graves and colleagues^[107] in a meta-analysis of 11 case control studies of patients with Alzheimer's disease observed a trend towards a decreasing risk of developing the disease with an increasing tobacco consumption ($p < 0.0003$). In a population based study of 152 patients with Alzheimer's disease and 180 matched controls,^[108] a smoking history was associated with an adjusted odds ratio of 0.61, although the greatest risk reduction was evident in those who smoked the least.

There is clearly a need for more epidemiological data on this subject, especially if this can incorporate neuropathological findings at autopsy. In a preliminary survey of normal elderly individuals, mean cortical plaque counts were lower in smokers compared with non- or ex-smokers.^[109] If a direct interaction between nicotine and the development of Alzheimer's disease is a valid interpretation of the various data so far, then the ability of nicotine to elevate nicotinic receptor numbers may be relevant. This is particularly so in areas such as the entorhinal cortex that have relatively high receptor binding density and a tendency to develop Alzheimer's pathology (sections 1.1 and 2.2). Furthermore, the loss of the urge to smoke with increasing

age may even be diagnostic of the development of Alzheimer's disease, with its concomitant decline in nicotinic receptors.

Maintaining normal levels of the receptor may be a useful long term strategy for brain survival, although tobacco smoking can hardly be advocated for other systemic aspects of survival. This raises the more philosophical question of whether morbidity of brain or body and associated mortality is the more acceptable or the less unacceptable alternative.

3.3 Therapeutic Trials of Nicotine

3.3.1 Effects in Parkinson's and Alzheimer's Diseases

Although nicotine has long been known to reduce tremor in patients with Parkinson's disease,^[110] the success of levodopa in symptomatic treatment has overshadowed the clinical use of nicotine in this disorder. By contrast, the absence of effective treatments for Alzheimer's disease (with the exception of anticholinesterases, which are moderately beneficial in a proportion of patients) has prompted the testing of numerous agents including nicotine. The rationale for the use of nicotine in Alzheimer's disease is persuasive. It is based on the degeneration or dysfunction of the cortical cholinergic projections and loss or down-regulation of nicotine receptors. The number of nicotinic receptors is increased by nicotine. The first report of beneficial effects of nicotine in Alzheimer's disease appeared in 1986.^[111] Following intravenous administration of nicotine (optimal dose 0.25 µg/kg/min) to 6 patients, intrusion errors in cognitive tests of free recall and category retrieval decreased, although there were also increases in anxiety and depressive affect.^[112]

According to Sahakian et al.^[113] subcutaneous administration of nicotine 0.4 to 0.8mg produced significant and marked improvement in discriminative sensitivity and reaction times in 7 patients with Alzheimer's disease. Effects were related to the role of cortical acetylcholine in modulating vigilance and discrimination. However, it is equally likely that stimulation of the relatively dense pop-

ulation of nicotinic receptors in thalamic nuclei, particularly lateral geniculate nucleus,^[36] would increase visual perception and attention.

In a more recent trial^[114] of single subcutaneous doses of nicotine (0.4 to 0.8mg), improvements in visual attention, reaction time and perception, but not memory *per se*, were evident in 22 patients with Alzheimer's disease and also in age-matched controls. Since Alzheimer's disease is primarily a disease affecting memory, it might be difficult based on these studies to consider nicotine as anything more than an adjunct to symptomatic therapy. It would act by exerting arousal effects that could be equally applicable to the normal aged population.

This does not diminish the potential value of nicotinic therapy in slowing neurodegeneration or its use in conjunction with muscarinic stimulation (activation of both types of cholinergic receptors is likely to be relevant in learning). The development of other nicotinic compounds that may act on disturbed memory mechanisms by selective targeting of receptor subtypes in the hippocampus or cortex may also be useful. Moreover, there is increasing evidence that nicotine enhances and antagonists such as mecamylamine diminish learning in healthy human volunteers and experimental animals.^[115] Furthermore, repeated administration of nicotine reverses memory deficits caused by fimbria lesions in rodents.^[116] Taken together these data suggest that the direct effect of nicotine on memory in patients with Alzheimer's disease should not be discounted in future trials.

3.3.2 Mechanism of Action

For over 3 decades research using experimental animals has indicated that the nicotinic antagonist mecamylamine impairs memory and other aspects of cognitive performance. More recently improvements in tasks involving attention in healthy human individuals have generally, although not always, been demonstrated after administration of nicotine.^[115,117]

The mechanisms of action of these drugs is likely to relate to the stimulatory effects of nicotine on the presynaptic release of neurotransmitters

such as acetylcholine, dopamine, noradrenaline (nor-epinephrine), serotonin (5-hydroxytryptamine; 5-HT) and γ -aminobutyric acid (GABA) [reviewed by Levin et al.^[117]]. Postsynaptic effects on specific transmitter systems are also evident. Nicotine has, for example, also been shown to enhance the amplitude of excitatory potentials mediated by glutamate in cortical pyramidal neurons.^[118]

Whether the increased levels of nicotine receptors evident in both humans and experimental animals exposed to tobacco or nicotine (see section 1.2) is the cause or consequence of the drug effect remains to be determined. Of particular importance in assessing the cognitive effects of nicotinic drugs is the increasing evidence that combined nicotinic and muscarinic drug effects may be additive.^[117] Both receptor types are present to varying degrees in both cortical and subcortical regions. Therefore, it is to be expected that the action of acetylcholine – and indeed effective cholinergic therapy in Alzheimer's disease for example – will be mediated through both muscarinic and nicotinic receptors.

3.3.3 Smoking and Psychiatric Disorders

At present, there are social trends to try and relegate the use of tobacco to the realms of unacceptable drug abuse. However, before this occurs it may be worth examining the reasons for smoking in apparently normal individuals. The mesolimbic dopaminergic system is considered to be involved in reinforcement or reward mechanisms. The perceived need of some individuals to smoke may relate to intrinsic variations from the norm in the physiology of this pathway, with increased well-being associated with nicotine stimulation. It is curious that there is a disproportionately high incidence of smoking in patients with schizophrenia or depression compared with the incidence in apparently mentally normal individuals.^[119,120] In schizophrenia, nicotine may offset the detrimental effects of conventional antipsychotics in blocking dopaminergic D₂ receptors, overstimulation of which can lead to tardive dyskinesia.

The depressed state is associated with a reduced latency and increased intensity of REM sleep. This

suggests an overactivity of the pedunculopontine cholinergic neurons that trigger this phenomenon. These neurons project principally to the thalamus, which has a high concentration of nicotinic receptors. Therefore, smoking in depression may involve a beneficial effect of nicotine on thalamic function. Interestingly, although nicotine increases receptor density in cortical areas and the hippocampus, thalamic receptors are relatively unaffected.^[27] This anatomical variation may account for a further nicotinic paradox, i.e. that nicotine exerts positive effects in Alzheimer's disease and depression. These disorders are associated with hypo- and hypercholinergic function, respectively, in basal forebrain and brain stem cholinergic pathways. Any programme aimed at developing new nicotinic compounds for the treatment of dementia may produce, as a by-product, useful therapy for other mental disorders.

Whatever the objectives in future nicotinic drug development, attention needs to be given to several potential problems. These include addiction, cardiovascular effects, fetal damage and, in differential dementia diagnosis, the fact that in multi-infarct dementia improved cognition is associated with cessation of tobacco smoking.^[121]

3.3.3 Mode of Administration

One of the great difficulties in trials of nicotinic agents is to select an appropriate mode of administration. This is because nicotinic receptors undergo rapid desensitisation and upregulation that may or may not be associated with normal function (section 1.2). Single doses administered intravenously or subcutaneously may be less appropriate than the 'bolus' mode of delivery associated with smoking. Transdermal nicotine patches although convenient may also be less effective than, for example, periodic inhalation. Nicotine inhalers are already being employed to induce smoking cessation and may be the most appropriate mode of delivery for long term therapy of CNS disorders. Alternatively, nicotinic agents that do not induce an increase in receptor number [such as lobeline or ABT 418 (see section 4)] may be more useful.

4. Spectrum of Nicotinic Compounds

The pharmacology of nicotinic receptor subtypes found at the neuromuscular junction and the CNS and within the CNS differs (section 1.2). It is also clear that the known nicotinic compounds are more efficacious at some receptor subtypes than others. In addition, studies in experimental animals indicate further differences between nicotinic agonists. For example, the nicotinic agonist lobeline has an only slightly lower affinity for brain nAChRs than nicotine,^[122] is more readily taken up by the brain than nicotine,^[123] and has been shown to be able to improve performance of learning and memory tasks in rats in a similar way to nicotine.^[124] However, exposure to lobeline *in vivo* does not cause upregulation of CNS nAChRs. This may also be the case for the novel compound ABT 418 [(*s*)-3-methyl-5-(1-methyl-2-pyrrolidinyl isoxazole)] (personal communication) (fig. 1).

Also, the activity of the nicotinic agonist cytosine does not consistently parallel that of nicotine. Results from oocytes expressing rat nAChR subunits indicate that cytosine may be a partial agonist at the $\alpha_4\beta_2$ receptor subtype, having little intrinsic nicotinic activity alone and inhibiting the activity of acetylcholine.^[125] Using rubidium flux from mouse fibroblasts stably transfected with chick $\alpha_4\beta_2$ receptors (as a measure of cation channel activity) and dopamine release from mouse striatal synaptosomes, cytosine has been shown to be roughly equipotent with (–)-nicotine, despite a greater affinity for the brain nAChRs than nicotine.^[40,126] In addition, cytosine was less effective than other agonists in promoting rubidium efflux from mouse brain synaptosomes.^[127] Also, unlike lobeline and (–)-nicotine, cytosine did not exhibit anxiolytic effects on the elevated plus-maze test in mice.^[128]

The ideal profile of a nicotinic agent for use in CNS disorders is not completely certain and would depend on its potential application. Clearly, brain specificity would be required. In view of the pharmacological differences between neuronal and neuromuscular nicotinic receptors, it would seem advan-

tageous to produce such compounds directed to a particular brain nAChR subtype. There is also a need to understand the significance of agonist-stimulated nAChR upregulation. There may be advantages to drugs with multiple sites of action, e.g. compounds with combined nicotinic and muscarinic effects (see section 3.3). Some muscarinic compounds do have nicotinic activity, e.g. arecoline.^[129] In addition, lobeline has an atropine-like effect in the gut^[130] and can inhibit dopamine reuptake.^[131] This latter effect could possibly be of benefit in protection against MPTP-like toxins.

5. Models for the Assessment of Nicotinic Drugs

Assessment of the selectivity and potency of nicotinic compounds has been made by measurement of binding affinity to brain nicotinic receptors. This is performed by measuring the displacement of radiolabelled nicotinic ligands (such as [³H]nicotine and other tritiated agonists for $\alpha_4\beta_2$ receptors and ¹²⁵I- α -bungarotoxin for α_7 receptors) by the compound in question from brain membrane preparations.^[132,133] Studies are worth extending to autoradiography (i.e. measuring displacement of radioligands in sections of tissue) so that possible site specificity within brain areas can be explored. However, equilibrium binding studies are believed to measure desensitised receptors, which have a higher affinity than receptors in the active state. Furthermore, it is unlikely that the affinity of agents for the 2 states has a consistent relationship. Therefore, functional models of the receptors are also required to assess compounds.

Many studies of receptor function have used cation flux from clonal cell lines to explore nicotine receptor channel opening.^[134-136] However, these cells tend to contain multiple α and β receptor subunits (e.g. PC12 cells^[11]) and, to date, none has been shown to contain the major brain subtype.

Nicotinic receptor subtypes expressed in xenopus oocytes can be used to assess nicotinic currents using patch clamping techniques.^[11] In addition, the chick $\alpha_4\beta_2$ receptor subtype has been stably transfected into mouse fibroblasts,^[137] and

rubidium flux in these cells has been used to assess nicotinic activity.^[126,138] These cells have the limitation of not possessing human receptor subtypes, which may be significant in view of the species variation that has been demonstrated with chick and rat subtype combinations expressed in oocytes.^[11] In addition, being a non-neuronal cell, there are limitations for their use in the study of events downstream from nicotinic activation and the control of receptor expression.

Brain tissue preparations *in vitro* have also been extensively employed to measure nicotinic activity, e.g. radiolabelled transmitter release or rubidium flux from nerve ending particle (synaptosome) preparations^[38-41,127] and whole cell recording from tissue slices.^[6,7,136]

All of these model systems have advantages in terms of relative ease of use and interpretation of results. However, they largely avoid the real complications of the *in vivo* situation. Such complications include the presence of the blood-brain barrier, drug metabolism and the inordinate complexity of potential interactions between different neurotransmitters, cells and tissues. Hence, many psychopharmacologists believe that proper tests can only be carried out *in vivo*, albeit generally confined to experimental animals prior to the final stage of drug development.

6. Conclusion

The brain nicotinic receptors represent potential therapeutic targets in dementia and possibly in ischaemia of the newborn. In human brain, high affinity binding sites for (–)-nicotine (presumed to be composed of $\alpha_4\beta_2$ subunits) are highly expressed in neonate brains, but there are also high levels of these receptors in areas of early Alzheimer's disease pathology. The loss of this receptor observed in normal aging, and to a greater extent in diseases of the elderly associated with dementia, may occur at an early stage of neurodegeneration and precede cell loss.

Upregulation of the number of these receptors occurs after long term exposure to nicotinic agonists and tobacco smoke. Although the precise re-

lationship between this process and receptor desensitisation is uncertain, there are indications that smoking tobacco may offer a degree of long term protection against Parkinson's disease and Alzheimer's disease. Acute exposure to (–)-nicotine enhances attention and information processing in patients with Alzheimer's disease.

Acknowledgements

We would like to thank Stephen Lloyd for his extensive literature search, and Maureen Middlemist and Dawn Hinds for their excellent secretarial help.

References

- Heinemann S, Boulter J, Connolly J, et al. The nicotinic receptor genes. *Clin Neuropharmacol* 1991; 14: S45-61
- Deneris ES, Connolly J, Rogers SW, et al. Pharmacological and functional diversity of neuronal nicotinic acetylcholine receptors. *Trends Pharmacol Sci* 1991; 12: 34-40
- Anand R, Conroy WG, Schoepfer R, et al. Neuronal nicotinic acetylcholine receptors expressed in *Xenopus* oocytes have a pentameric quaternary structure. *J Biol Chem* 1991; 266: 11192-8
- Clarke PBS. The fall and rise of neuronal α -bungarotoxin binding proteins. *Trends Pharmacol Sci* 1992; 13: 407-13
- Wonnacott S. Alpha-bungarotoxin binds to low-affinity nicotine binding sites in rat brain. *J Neurochem* 1986; 47: 1706-12
- de la Garza R, Bickford-Wimer PC, Hoffer BJ, et al. Heterogeneity of nicotine actions in the rat cerebellum: an in vivo electrophysiologic study. *J Pharmacol Exp Ther* 1987; 240: 689-95
- de la Garza R, Freedman R, Hoffer BJ. α -Bungarotoxin blockade of nicotine electrophysiological actions in cerebellar Purkinje neurons. *Neurosci Lett* 1989; 99: 95-100
- Mulle C, Vidal C, Benoit P, et al. Existence of different subtypes of nicotinic acetylcholine receptors in the rat habenulo-interpeduncular system. *J Neurosci* 1991; 11: 2588-97
- Sargent PB. The diversity of neuronal nicotinic acetylcholine receptors. *Ann Rev Neurosci* 1993; 16: 403-43
- Sugaya K, Giacobini E, Chiappinelli VA. Nicotinic acetylcholine receptor subtypes in frontal cortex: changes in Alzheimer's disease. *J Neurosci Res* 1990; 27: 349-59
- Role LW. Diversity in primary structure and function of neuronal nicotinic acetylcholine receptor channels. *Curr Opin Neurobiol* 1992; 2: 254-64
- Martino-Barrows AM, Kellar KJ. [3 H]Acetylcholine and [3 H](–)nicotine label the same recognition site in rat brain. *Mol Pharmacol* 1987; 31: 169-74
- Marks MJ, Stitzel JA, Romm E, et al. Nicotinic binding sites in rat and mouse brain: comparison of acetylcholine, nicotine and alpha-bungarotoxin. *Mol Pharmacol* 1986; 30: 427-36
- Clarke PBS, Schwartz RD, Paul SM, et al. Nicotinic binding in rat brain: autoradiographic comparison of [3 H]acetylcholine, [3 H]nicotine, and [125 I] α -bungarotoxin. *J Neurosci* 1985; 5: 1307-15
- Härfstrand A, Adem A, Fuxe F, et al. Distribution of nicotinic cholinergic receptors in the rat tel- and diencephalon: a quantitative receptor autoradiographical study using [3 H]-acetylcholine, [α^{125} I]bungarotoxin and [3 H]nicotine. *Acta Physiol Scand* 1988; 132: 1-14
- Schoepfer R, Whiting P, Esch F, et al. cDNA clones coding for the structural subunit of a chicken brain nicotinic acetylcholine receptor. *Neuron* 1988; 1: 241-8
- Whiting PJ, Esch F, Shimasaki S, et al. Neuronal nicotinic acetylcholine receptor β subunit is coded by the cDNA $\alpha 4$. *FEBS Lett* 1987; 219: 459-63
- Nakayama H, Nakashima T, Kuroguchi Y. $\alpha 4$ is a major acetylcholine binding subunit of cholinergic ligand affinity – purified nicotinic acetylcholine receptor from rat brains. *Neurosci Lett* 1991; 121: 122-4
- Flores CM, Rogers SW, Pabreza LA, et al. A subtype of nicotinic cholinergic receptor in rat brain is composed of $\alpha 4 \beta 2$ subunits and is upregulated by chronic nicotine treatment. *Mol Pharmacol* 1992; 41: 31-7
- Luetje CW, Patrick J. Both α - and β -subunits contribute to the agonist sensitivity of neuronal nicotine acetylcholine receptors. *J Neurosci* 1991; 11: 837-45
- Connolly J, Boulter J, Heinemann SF. $\alpha 4 \beta 2$ and other nicotinic acetylcholine receptor subtypes as targets of psychoactive and addictive drugs. *Br J Pharmacol* 1992; 105: 657-66
- Couturier S, Bertrand D, Matter J, et al. A neuronal nicotinic acetylcholine receptor subunit ($\alpha 7$) is developmentally regulated and forms a homo-oligomeric channel blocked by α BT. *Neuron* 1990; 5: 847-56
- Bertrand D, Bertrand S, Ballivet M. Pharmacological properties of the homomeric $\alpha 7$ receptor. *Neurosci Lett* 1992; 146: 87-90
- Séguéla P, Wadiche J, Dineley-Miller K, et al. Molecular cloning, functional properties, and distribution of rat brain $\alpha 7$: a nicotinic cation channel highly permeable to calcium. *J Neurosci* 1993; 13: 596-604
- Gerzanich V, Anand R, Lindstrom J. Homomers of $\alpha 8$ and $\alpha 7$ subunits of nicotinic receptors exhibit similar channel but contrasting binding site properties. *Mol Pharmacol* 1994; 45: 212-20
- Keyser KT, Britto LRG, Schoepfer R, et al. Three subtypes of α -bungarotoxin-sensitive nicotinic acetylcholine receptors are expressed in chick brain. *J Neurosci* 1993; 13: 442-54
- Pauly JR, Marks MJ, Gross SD, et al. An autoradiographic analysis of cholinergic receptors in mouse brain after chronic nicotine treatment. *J Pharmacol Exp Ther* 1991; 258: 1127-36
- Marks MJ, Campbell SM, Romm E, et al. Genotype influences the development of tolerance to nicotine in the mouse. *J Pharmacol Exp Ther* 1991; 259: 392-402
- Marks MJ, Pauly JR, Gross SD, et al. Nicotine binding and nicotinic receptor subunit RNA after chronic nicotine treatment. *J Neurosci* 1992; 12: 2765-84
- Wada E, Wada K, Boulter J, et al. Distribution of alpha 2, alpha 3, alpha 4, and beta 2 neuronal nicotinic receptor subunit mRNAs in the central nervous system: a hybridization histochemical study in the rat. *J Comp Neurol* 1989; 284: 314-35
- Wada E, McKinnon D, Heinemann S, et al. The distribution of mRNA encoded by a new family member of the neuronal nicotinic acetylcholine receptor gene family ($\alpha 5$) in the rat central nervous system. *Brain Res* 1990; 526: 45-53
- Dineley-Miller K, Patrick J. Gene transcripts for the nicotinic acetylcholine receptor subunit, beta4 are distributed in multiple areas of the rat central nervous system. *Mol Brain Res* 1992; 16: 339-44
- Perry EK, Smith CJ, Perry RH, et al. Nicotinic (3 H-nicotine) receptor binding in human brain: characterization and involvement in cholinergic neuropathology. *Neurosci Res Comm* 1989; 5: 117-24

34. Perry EK, Smith CJ, Perry RH, et al. Regional distribution of muscarinic and nicotinic cholinergic receptor binding in the human brain. *J Chem Neuroanat* 1989; 2: 189-99
35. Xuereb JH, Perry EK, Candy JM, et al. Parameters of cholinergic neurotransmission in the thalamus in Parkinson's disease and Alzheimer's disease. *J Neurol Sci* 1990; 99: 185-97
36. Court JA, Perry EK. Distribution of nicotinic receptors in the CNS. In: Stone TW, editor. *Acetylcholine*. Boca Raton, Florida: CRC Press Inc. In press
37. Perry EK, Court JA, Johnson M, et al. Autoradiographic distribution of [3 H]nicotine binding in human cortex: relative abundance in subicular complex. *J Clin Anat* 1992; 5: 399-405
38. Rapier C, Lunt GG, Wonnacott S. Stereoselective nicotine-induced release of dopamine from striatal synaptosomes: concentration dependence and repetitive stimulation. *J Neurochem* 1988; 50: 1123-30
39. Rapier C, Lunt GG, Wonnacott S. Nicotinic modulation of [3 H]dopamine release from striatal synaptosomes: pharmacological characterisation. *J Neurochem* 1990; 54: 937-45
40. Grady S, Marks MM, Wonnacott S, et al. Characterization of nicotinic receptor-mediated [3 H]dopamine release from synaptosomes prepared from mouse striatum. *J Neurochem* 1992; 59: 848-56
41. Rowell PP, Winkler DL. Nicotinic stimulation of [3 H]acetylcholine release from mouse cerebral cortical synaptosomes. *J Neurochem* 1984; 43: 1593-8
42. Smith CJ, Court JA, Keith AB, et al. Increases in muscarinic stimulated hydrolysis of inositol phospholipids in rat hippocampus following cholinergic deafferentation are not paralleled by alterations in cholinergic receptor density. *Brain Res* 1989; 485: 317-24
43. Miyai I, Veno S, Yorifuji S, et al. Alterations in neocortical expression of nicotinic acetylcholine receptor in mRNAs following unilateral lesions of the rat nucleus basalis magnocellularis. *J Neurol Transm* 1990; 82: 79-91
44. Sahin M, Bowen WD, Donoghue JP. Location of nicotinic and muscarinic cholinergic and μ -opioid receptors in rat cerebral neocortex: evidence from thalamic and cortical lesions. *Brain Res* 1992; 579: 135-47
45. Prusky GT, Shaw C, Cynader MS. Nicotine receptors are located on lateral geniculate nucleus terminals in cat visual cortex. *Brain Res* 1987; 412: 131-8
46. Schröder H, Zilles K, Maelicke A, et al. Immunohisto- and cytochemical localization of cortical nicotinic cholinergic receptors in rat and man. *Brain Res* 1989; 502: 287-95
47. Hill JA, Zoli M, Bourgeois J-P, et al. Immunocytochemical localization of a neuronal nicotinic receptor: β_2 -subunit. *J Neurosci* 1993; 13: 1551-68
48. Afar R, Clarke PBS, Goldstein G, et al. Thymopoietin, a polypeptide ligand for the α -bungarotoxin binding site in brain: an autoradiographic study. *Neurosci* 1992; 48: 641-53
49. Figl A, Cohen BN, Quick MW, et al. Regions of beta 4, beta 2 subunit chimeras that contribute to the agonist selectivity of neuronal nicotinic receptors. *FEBS Lett* 1992; 308: 245-8
50. Cachelin AB, Jaggi R. Beta subunits determine the time course of desensitization in rat alpha 3 neuronal nicotinic acetylcholine receptors. *Pflügers Arch* 1991; 419: 579-82
51. Mulle C, Choquet D, Korn H, et al. Calcium influx through nicotinic receptor in rat central neurons: its relevance to cellular regulations. *Neuron* 1992; 8: 135-43
52. Vernino S, Amador M, Leutje CW, et al. Calcium modulation and high calcium permeability of neuronal nicotinic acetylcholine receptors. *Neuron* 1992; 8: 127-34
53. Trouslard J, Mirsky R, Jessen KR, et al. Intracellular calcium changes associated with cholinergic nicotinic receptor activation in cultivated myenteric plexus neurones. *Brain Res* 1993; 624: 103-8
54. Trouslard J, Marsh SJ, Brown DA. Calcium entry through nicotinic receptor channels and calcium channels in cultured superior cervical ganglion cells. *J Physiol* 1993; 468: 53-71
55. Gerzanich V, Anand R, Lindstrom J. Homomers of $\alpha 8$ and $\alpha 7$ subunits of nicotinic receptors exhibit similar channel but contrasting binding site properties. *Mol Pharmacol* 1994; 45: 212-20
56. Sando SB, Costa ACS, Patrick JW. Barium permeability of neuronal nicotinic receptor $\alpha 7$ expressed in *Xenopus* oocytes. *Biophys J* 1993; 65: 2614-21
57. Lipton SA, Kater SB. Neurotransmitter regulation of neuronal outgrowth, plasticity and survival. *Trends Neurosci* 1989; 12: 265-70
58. Zheng JQ, Felder M, Connor JA, et al. Turning of nerve growth cones induced by neurotransmitters. *Nature* 1994; 368: 140-4
59. Yang X-H, Buccafusco JT. Comparison of the ability of direct and indirect-acting nicotinic agonists to induce nicotinic receptor up-regulation in rat cerebral cortex. *Drug Dev Res* 1994; 31: 95-101
60. Bhat RV, Turner SL, Selvaag SR, et al. Regulation of brain nicotinic receptors by chronic agonist infusion. *J Neurochem* 1991; 56: 1932-9
61. Sanderson EM, Drasdo AL, McCrea K, et al. Upregulation of nicotinic receptors following continuous infusion of nicotine is brain-region-specific. *Brain Res* 1993; 617: 349-52
62. Benwell MEM, Balfour DJK, Anderson JM. Evidence that tobacco smoking increases the density of (-)-[3 H]nicotine binding sites in human brain. *J Neurochem* 1988; 50: 1243-7
63. Wonnacott S. The paradox of nicotinic acetylcholine receptor upregulation by nicotine. *Trend Pharmacol Sci* 1990; 11: 216-9
64. Mark MJ, Grady SR, Collins AC. Downregulation of nicotinic receptor function and chronic nicotine infusion. *J Pharmacol Exp Ther* 1993; 266: 1268-79
65. Rowell PP, Wonnacott S. Evidence for functional activity of up-regulated nicotine binding sites in rat striatal synaptosomes. *J Neurochem* 1990; 55: 2105-10
66. Collins AC, Romm E, Wehner JM. Dissociation of the apparent relationship between nicotine tolerance and up-regulation of nicotinic receptors. *Brain Res Bull* 1990; 25: 373-9
67. Eimerl S, Schramm M. The quantity of calcium that appears to induce neuronal death. *J Neurochem* 1994; 62: 1223-6
68. Kinney HC, O'Donnell TY, Kringer P, et al. Early developmental changes in (3 H)nicotine binding in the human brainstem. *Neuroscience* 1993; 55: 1127-38
69. Court JA, Piggott MA, Perry EK, et al. Age associated decline in high affinity nicotinic binding in human brain frontal cortex does not correlate with changes in choline acetyltransferase activity. *Neurosci Res Comm* 1992; 10: 125-33
70. Navarro HA, Seidler FJ, Eyler JP, et al. Effects of prenatal exposure on development of central and peripheral cholinergic neurotransmitter systems. *J Pharmacol Exp Ther* 1989; 251: 894-900
71. Flynn DD, Mash DC. Characterization of L-(3 H)nicotine binding in human cerebral cortex: comparison between Alzheimer's disease and the normal. *J Neurochem* 1986; 47: 1948-54
72. Schröder H, Giacobini E, Struble RG, et al. Nicotinic cholinergic neurons in the frontal cortex are reduced in Alzheimer's disease. *Neurobiol Aging* 1991; 12: 259-62

73. Araujo DM, Lapchak PA, Robitaille Y, et al. Differential alteration of various cholinergic markers in cortical and subcortical regions of human brain in Alzheimer's disease. *J Neurochem* 1988; 50: 1914-23
74. Aubert I, Araujo DM, Cécylre D, et al. Comparative alterations of nicotine and muscarinic binding sites in Alzheimer's disease and Parkinson's disease. *J Neurochem* 1992; 58: 529-41
75. London ED, Ball MJ, Waller SB. Nicotinic binding sites in cerebral cortex and hippocampus in Alzheimer's dementia. *Neurochem Res* 1989; 14: 745-50
76. Nordberg A, Winblad B. Reduced number of ^3H -nicotine and ^3H -acetylcholine binding sites in the frontal cortex of Alzheimer brains. *Neurosci Lett* 1986; 72: 115-9
77. Nordberg A, Adem A, Hardy J, et al. Change in nicotinic receptor subtypes in temporal cortex of Alzheimer brains. *Neurosci Lett* 1988; 86: 317-21
78. Perry EK, Perry RH, Smith CJ, et al. Cholinergic receptors in cognitive disorders. *Can J Neurol Sci* 1986; 13: 521-7
79. Perry EK, Smith CJ, Court JA, et al. Cholinergic nicotinic and muscarinic receptors in dementia of Alzheimer, Parkinson and Lewy body types. *J Neural Transm Park Dis Dement Sect* 1990; 2: 149-58
80. Rinne JO, Myllykylä T, Lonnberg P, et al. A postmortem study of brain nicotinic receptors in Parkinson's and Alzheimer's disease. *Brain Res* 1991; 547: 167-70
81. Shimohama S, Taniguchi T, Fujiwara H, et al. Changes in nicotinic and muscarinic cholinergic receptors in Alzheimer type dementia. *J Neurochem* 1986; 46: 288-93
82. Sugaya K, Giacobini E, Chiappinelli VA. Nicotinic acetylcholine receptor subtypes in human frontal cortex: changes in Alzheimer's disease. *J Neurosci Res* 1990; 27: 349-59
83. Whitehouse PJ, Martino AM, Antuono PG, et al. Nicotinic acetylcholine binding sites in Alzheimer's disease. *Brain Res* 1986; 371: 146-51
84. Perry EK, Johnson M, Kerwin JM, et al. Convergent archicortical cholinergic activities in aging and Alzheimer's disease. *Neurobiol Aging* 1992; 13: 393-400
85. Wagstaff AJ, McTavish D. Tacrine: a review of its pharmacodynamic and pharmacokinetic properties, and therapeutic efficacy in Alzheimer's disease. *Drugs Aging* 1994; 4: 510-40
86. Lange KW, Wells FR, Jenner P, et al. Altered muscarinic and nicotinic receptor densities in cortical and subcortical regions in Parkinson's disease. *J Neurochem* 1993; 60: 197-203
87. Moll H. The treatment of post-encephalitic Parkinsonism by nicotine. *BMJ* 1926; 1: 1079-81
88. Nefzger MD, Quadfasel FA, Karl VC. A retrospective study of smoking in Parkinson's disease. *Am J Epidemiol* 1968; 88: 149-58
89. Baumann RJ, Jameson HD, McKean HE, et al. Cigarette smoking and Parkinson's disease. 1. Comparison of cases with matched neighbours. *Neurology* 1980; 30: 839-43
90. Martilla RJ, Rinne UK. Smoking and Parkinson's disease. *Acta Neurol Scand* 1980; 62: 322-5
91. Godwin-Austen RB, Lee PN, Marmot MG, et al. Smoking and Parkinson's disease. *J Neurol Neurosurg Psychiatry* 1982; 45: 577-81
92. Cazzato G, Capus L, Monti, et al. Smoking and Parkinson's disease. *Riv Neurol* 1985; 55: 79-87
93. Shahi GS, Moomchala SM. Smoking and Parkinson's disease – a new perspective. *Rev Environ Health* 1991; 9: 123-36
94. Paulson GW, Dadmehr N. Is there a premorbid personality typical for Parkinson's disease? *Neurology* 1991; 41 5 Suppl. 2: 73-6
95. Paulson GW. Addiction to nicotine is due to high intrinsic levels of dopamine. *Med Hypotheses* 1992; 38: 206-7
96. Hedin CA. Smoker's melanosis may explain the lower hearing loss and lower frequency of Parkinson's disease found among tobacco smokers – a new hypotheses. *Med Hypotheses* 1991; 35: 247-9
97. Gresham LS, Molgaard CA, Smith RA. Induction of cytochrome P-450 enzymes via tobacco smoke: a potential mechanism for developing resistance to environmental toxins as related to parkinsonism and other neurologic diseases. *Neuro-epidemiology* 1993; 12: 114-6
98. Yong VW, Perry TL. Monoamine oxidase B, smoking and Parkinson's disease. *J Neurol Sci* 1986; 72: 265-72
99. Baron JA. Cigarette smoking and Parkinson's disease. *Neurology* 1986; 36: 1490-6
100. Janson AM, Fuxe K, Agnati LF, et al. Chronic nicotine treatment counteracts disappearance of tyrosine-hydroxylase-immunoreactive nerve cell bodies, dendrites and terminals in the mesostriatal dopamine system of the male rat after partial hemitransection. *Brain Res* 1988; 455: 332-45
101. Janson AM, Meana JJ, Goñy M, et al. Chronic nicotinic treatment counteracts the decrease in extracellular neostriatal dopamine induced by a unilateral transection at the mesodiencephalic junction in rats: a microdialysis study. *Neurosci Lett* 1991; 134: 88-92
102. Janson AM, Fuxe F, Sundstrom E, et al. Chronic nicotine treatment partly protects against the 1-methyl-4-phenyl-2,3,6-tetrahydropyridine-induced degeneration of nigrostriatal dopamine neurons in the black mouse. *Acta Physiol Scand* 1988; 132: 589-91
103. Sjak-Shie NN, Meyer EM. Effects of chronic nicotine and pilocarpine administration on neocortical neuronal density and (^3H) GABA uptake in nucleus basalis lesioned rats. *Brain Res* 1993; 624: 295-8
104. Ishikawa A, Miyatake T. Effects of smoking in patients with early onset Parkinson's disease. *J Neurol Sci* 1993; 117: 28-32
105. Shalat SL, Seltzer B, Pidcock C, et al. Risk factors for Alzheimer's disease: a case-control study. *Neurology* 1987; 37: 1630-3
106. van Duijn CM, Hofman A. Relation between nicotine intake and Alzheimer's disease. *BMJ* 1991; 302: 1491-4
107. Graves AB, van Duijn CM, Chandra V, et al. Alcohol and tobacco consumption as risk factors for Alzheimer's disease: a collaborative re-analysis of case-control studies. EU-RODEM risk factors Research Group. *Int J Epidemiol* 1991; 20 Suppl. 2: 48-57
108. Brenner DE, Kukull WA, van Belle G, et al. Relationship between cigarette smoking and Alzheimer's disease in a population-base case-control study. *Neurology* 1993; 43: 293-300
109. Perry EK, Perry RH. Neurochemical pathology and therapeutic strategies in degenerated dementia. *Int Rev Psychiatry* 1993; 5: 363-80
110. Marshall J, Schneiden H. Effect of adrenaline, noradrenaline, atropine and nicotine on some types of human tremor. *J Neurol Neurosurg Psychiatry* 1966; 29: 214-8
111. Newhouse PA, Sunderland T, Thompson K, et al. Intravenous nicotine in a patient with Alzheimer's disease. *Am J Psychiatry* 1986; 143: 1494-5
112. Newhouse PA, Sunderland T, Tariot PN, et al. Intravenous nicotine in Alzheimer's disease: a pilot study. *Psychopharmacology* 1988; 95: 171-5
113. Sahakian B, Jones G, Levy R, et al. The effects of nicotine on attention, information processing and short term memory in

- patients with dementia of the Alzheimer type. *Br J Psychiatry* 1989; 154: 797-800
114. Jones GM, Sahakian BJ, Levy R, et al. Effects of acute subcutaneous nicotine on attention, information processing and short term memory in Alzheimer's disease. *Psychopharmacology* 1992; 108: 485-94
115. Levin ED. Nicotinic systems and cognitive function. *Psychopharmacology* 1992; 108: 417-31
116. Levin ED, Christopher C, Briggs S, et al. Chronic nicotine reverses working memory deficits caused by lesions of the fimbria or medial basolateral projection. *Brain Res Cogn Brain Res* 1993; 1: 137-43
117. Levin ED, Rose JE, McGurk SR, et al. Characterization of the cognitive effects of combined muscarinic and nicotinic blockade. *Behav Neural Biol* 1990; 53: 103-12
118. Vidal C. Nicotinic potentiation of glutamatergic synapses in the prefrontal cortex: new insight into the analysis of the role of nicotinic receptors in cognitive functions. *Drug Dev Res* 1994; 31: 120-6
119. Newhouse PA, Hughes JR. The role of nicotine and nicotinic mechanisms in neuropsychiatric disease. *J Addict* 1991; 86: 521-6
120. Glassman AH. Cigarette smoking: implications for psychiatric illness. *Am J Psychiatry* 1993; 150: 546-53
121. Meyer JS, Judd BW, Tawakina T, et al. Improved cognition after control of risk factors for multi-infarct dementia. *JAMA* 1986; 256: 2203-9
122. Reavill C, Jenner P, Kumar R, et al. High affinity binding of [³H] (-)-nicotine to rat brain membranes and its inhibition by analogs of nicotine. *Neuropharmacology* 1988; 27: 235-41
123. Reavill C, Walthers B, Stolerman IP, et al. Behavioural and pharmacokinetic studies on nicotine, cytosine and lobeline. *Neuropharmacology* 1990; 29: 619-24
124. Decker MW, Majchrzak, Arneric SP. Effects of lobeline, a nicotinic receptor agonist, on learning and memory. *Pharmacol Biochem Behav* 1993; 45: 571-6
125. Papke RL, Heinemann SF. Partial agonists properties of cytosine on neuronal nicotinic receptors containing the β_2 subunit. *Mol Pharmacol* 1994; 45: 142-9
126. Court JA, Perry EK. Dementia: the neurochemical basis of putative transmitter orientated therapy. *Pharmac Ther* 1991; 52: 423-43
127. Marks MJ, Farnham DA, Grady SR, et al. Nicotinic receptor function determined by stimulation of rubidium efflux from mouse brain synaptosomes. *J Pharmacol Exp Ther* 1993; 264: 542-52
128. Brioni JD, O'Neill AB, Kim DJB, et al. Nicotinic receptor agonists exhibit anxiolytic-like effects on the elevated plus-maze test. *Eur J Pharmacol* 1993; 238: 1-8
129. Gilman AG, Goodman LS, Rall TW, et al. The pharmacological basis of therapeutics. New York: Macmillan, 1985
130. Barlow RB, Franks F. Specificity of some ganglionic stimulants. *Br J Pharmacol* 1971; 42: 137-42
131. Debler EA, Hashim A, Lajha A, et al. Ascorbic acid and striatal transport of [³H]1-methyl-4-phenylpyridine (MPP⁺) and [³H]dopamine. *Life Sci* 1988; 42: 2553-9
132. Wonnacott S, Jackman S, Swanson KL, et al. Nicotinic pharmacology of anatoxin analogs. II. Side chain structure - activity relationships at neural nicotinic ligand binding sites. *J Pharmacol Exp Ther* 1991; 259: 387-91
133. Abood LG, Salles KS, Maiti A. Structure-activity studies of carbamate and other esters: agonists and antagonists to nicotine. *Pharmacol Biochem Behav* 1988; 30: 403-8
134. Lukas RJ. Pharmacological distinctions between functional nicotinic acetylcholine receptors on the PC12 rat pheochromocytoma and the TE671 human medullo blastoma. *J Pharmacol Exp Ther* 1989; 251: 175-82
135. Lukas RJ, Norman SA, Lucero L. Characterizations of nicotinic acetylcholine receptors expressed by cells of the SH-SY5Y human neuroblastoma clonal cell line. *Mol Cell Neurosci* 1992; 4: 1-2
136. Lukas RJ, Bencherif M. Heterogeneity and regulation of nicotinic acetylcholine receptors. *Int Rev Neurobiol* 1992; 34: 25-131
137. Whiting P, Schoepfer R, Lindstrom J, et al. Structural and pharmacological characterization of the major brain nicotinic acetylcholine receptor subtype stably expressed in mouse fibroblasts. *Mol Pharmacol* 1991; 40: 463-72
138. Thomas P, Stephens M, Wilkie G, et al. (+)-Anatoxin-a is a potent agonist at neuronal nicotinic acetylcholine receptors. *J Neurochem* 1993; 60: 2308-11

Correspondence and reprints: Dr Jennifer A. Court, MRC Neurochemical Pathology Unit, Newcastle General Hospital, Westgate Road, Newcastle upon Tyne, NE4 6BE, England.